

Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	FinRelief - AI
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>React.js, Vite, HTML, CSS, JavaScript</i>	<i>Builds a responsive, interactive, and reusable component-based user interface with fast development and optimized production builds.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, FastAPI, Uvicorn</i>	Provides high-performance REST APIs, financial processing, input validation, authentication, and AI integration.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>SQLite</i>	<i>Provides lightweight relational storage for user accounts, loan details, financial profiles, and application records.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Vite Production Build, Uvicorn ASGI Server</i>	<i>Prepares the React frontend and FastAPI backend for efficient production deployment and scalable server execution.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Enables source code version tracking, organized repository management, backup, collaboration, and branch-based development.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Google Gemini API, JWT, Werkzeug</i>	<i>Enables AI-powered negotiation strategies and letter generation, secure token-based authentication, and password hashing</i>